

Quasi-complete graphs need not maximize spectral moments

Disproof of Conjecture 00000003725

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Moment convention and scope. For a finite simple graph G with actual adjacency matrix A_G , the standard adjacency spectral moment of order k is

$$M_k(G) = \sum_j \lambda_j(A_G)^k = \text{tr}(A_G^k).$$

The trace equality follows by diagonalizing the real symmetric adjacency matrix. It also counts length- k closed walks. We use $k = 4$ and compare actual graphs with both vertex count and edge count fixed. A single competing graph with larger fourth moment is enough to refute the asserted quasi-complete extremum. No lexicographic ordering of the entire moment sequence is stated in the problem, and we make no claim about that different ordering.

The prescribed quasi-complete graph. The standard quasi-complete construction writes $m = \binom{r}{2} + s$, $0 \leq s < r$: take a clique on r vertices, join one new vertex to s clique vertices, and leave any remaining vertices isolated. This convention is described in *Sum of squares of degrees in a graph*, Section 1, by Abrego, Fernandez-Merchant, Neubauer and Watkins (2009), <https://emis.de/ft/14223>. For $n = 5, m = 4$, the parameters are $r = 3, s = 1$. Thus $Q = QC(5, 4)$ has edges 01, 02, 12, 03 and isolated vertex 4. Compare it with the star $S = K_{1,4}$, whose edges are 01, 02, 03, 04. Both are actual simple five-vertex, four-edge graphs.

Exact fourth moments. Their adjacency matrices, in vertex order 0, 1, 2, 3, 4, are

$$A_Q = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad A_S = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Because the matrices are symmetric, $\text{tr}(A^4) = \sum_{i,j} (A_{ij}^2)^2$. For Q , the nonzero part of A_Q^2 is

$$\begin{pmatrix} 3 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix},$$

whose squared entries sum to $11 + 7 + 7 + 3 = 28$. For the star, A_S^2 consists of the central diagonal entry 4 and an all-one 4×4 leaf block, giving $4^2 + 16 = 32$. Consequently

$$M_4(S) = 32 > 28 = M_4(Q).$$

Hence $QC(5, 4)$ does not maximize the actual fourth spectral moment among graphs with the prescribed counts, and the conjecture is false. The same strict comparison holds for moments divided by the common vertex count five.

Formal certificate. Lean constructs the actual graphs and checks the canonical parameters and both edge sets. It verifies the real adjacency matrices and calculates the fourth-power traces. The final theorem disproves the asserted maximum. All calculations are checked by the Lean kernel; final audits use only standard Lean and Mathlib axioms.